

Project 2: Automation of Concrete Mix Design Mix Scenarios Research and Documentation

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Scenario #1 47B-3500

47B-3500 is a class of concrete that has a compressive strength of 3500 psi. The other concrete classes will follow this same structure, with the compressive strength coming at the end of the name (Nebraska Department of Transportation, n.d-a). This type of concrete is designed to meet the demands of construction projects such as roads, sidewalks, and infrastructure. Portland cement is used in this class of concrete (Nebraska Department of Transportation, n.d-b).

Scenario #2 Flowable Fill

Flowable fill concrete is a self-compacting and fast settling concrete used for backfilling voids, trenches, and structures. It eliminates the need for compaction equipment and reduces labor. The concrete flows into tight spaces and sets quickly (Flashfill Services, 2025). Flowable fill is a mixture of cement, fly ash, fine sand, water, and air (City of Lincoln, n.d.-a). The approximate compressive strength should range from 85 to 175 psi. No less than 95% of sand should pass through the No.4 sieve which is under the category of fine aggregate. No more than 5% should pass through the No.200 sieve which is the finest sieve.

Scenario #3 L-3500

L-3500 is a class of concrete that is a specific type of Portland Cement Concrete (PCC). Its general use is for pavements, sidewalks, and structures. This class of concrete is usually used in areas of high compressive strength being able to withstand significant loads (City of Lincoln, n.d.-b). The minimum PSI strength for this mixture is 3500 psi.

Scenario #4 LB-2750

LB-2750 is a class of concrete that is used for new construction of residential streets (City of Lincoln, n.d.-c). This is generally a pavement base with the minimum PSI being 2750.

Scenario #1 47B-3500 - Nebraska Department of Transportation. (2020). Standard specifications for highway construction			
water_cement_ratio_E	0.45	air_content_F	6%
Cementitious Materials		Specific Gravities	
cement_A	470.0 lb/yd ³	sg_cement_J	3.15
fly_ash_B	120.0 lb/yd ³	sg_fly_ash_K	2.40
Aggregate Percentages			
percent_fine_G	40.0%	sg_fine_N	2.65
percent_coarse_H	60.0%	sg_coarse_O	2.65

Scenario #2 Flowable Fill - City of Lincoln. (2020). Standard specifications for highway construction			
water_cement_ratio_E	0.45	air_content_F	10%
Cementitious Materials		Specific Gravities	
cement_A	60.0 lb/yd ³	sg_cement_J	3.15
fly_ash_B	200.0 lb/yd ³	sg_fly_ash_K	2.40
Aggregate Percentages			
percent_fine_G	100.0%	sg_fine_N	2.65

Scenario #3 L-3500 - City of Lincoln. (2020). Standard specifications for highway construction			
water_cement_ratio_E	0.5	air_content_F	6%
Cementitious Materials		Specific Gravities	
cement_A	564.0 lb/yd ³	sg_cement_J	3.15
Aggregate Percentages			
percent_fine_G	70.0%	sg_fine_N	2.65
percent_coarse_H	30.0%	sg_coarse_O	2.65

Scenario #4 LB-2750 - City of Lincoln. (2020). Standard specifications for highway construction			
water_cement_ratio_E	0.60	air_content_F	6%
Cementitious Materials		Specific Gravities	
cement_A	423.0 lb/yd ³	sg_cement_J	3.15
Aggregate Percentages			
percent_fine_G	60.0%	sg_fine_N	2.65
percent_coarse_H	40.0%	sg_coarse_O	2.65

Works Cited

2020 city of Lincoln Standard Specifications 300. (n.d.-a).

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